PULLBACK OF SHEAVES OF MODULES IS LEFT ADJOINT TO PUSHFORWARD AT THE SHEAF LEVEL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sheaves of f^{-1}O-modules correspond to sheafifications of presheaves of f_pO-modules (lemma-sheafification-presheaf-modules). invoked at proof lines 3176-3177. Provides the first equality between sheaf and presheaf module Hom sets.
- Presheaf-level push-pull adjunction for modules (lemma-adjoint-push-pull-presheaves-modules). invoked at proof lines 3177-3179. Provides the second equality.

Source reference. Original lemma 008X: sources/stacks-sheaves/pinned-original-source.tex, source lines 3113–3113. Cited at statement lines 3162-3163 to name the module structure on f^{-1}G; part of the asserted bijection.

Source reference. Original lemma 008W: sources/stacks-sheaves/pinned-original-source.tex, source lines 3099–3099. Cited at statement lines 3162-3163 to name the module structure on f_*F; part of the asserted bijection.

Source reference. Original lemma 008U: sources/stacks-sheaves/pinned-original-source.tex, source lines 2999–2999. Cited at proof lines 3177-3179 as the justification of the second equality; load-bearing inside the counted span.

Source reference. Original lemma 0089: sources/stacks-sheaves/pinned-original-source.tex, source lines 2042–2042. Cited at proof lines 3176-3177 as the justification of the first equality; load-bearing inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{O}$ be a sheaf of rings on Y. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}$-modules. Let $\mathcal{F}$ be a sheaf of $f^{-1}\mathcal{O}$-modules. Then*

$$\mathrm{Mor}_{Mod(f^{-1}\mathcal{O})}(f^{-1}\mathcal{G}, \mathcal{F}) = \mathrm{Mor}_{Mod(\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}).$$

Here we use Lemmas 008X and 008W, and we think of $f_*\mathcal{F}$ as an $\mathcal{O}$-module by restriction via $\mathcal{O} \to f_*f^{-1}\mathcal{O}$.

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Proof. Argue by the equalities

$$\begin{aligned} \mathrm{Mor}_{Mod(f^{-1}\mathcal{O})}(f^{-1}\mathcal{G}, \mathcal{F}) &= \mathrm{Mor}_{Mod(f_p\mathcal{O})}(f_p\mathcal{G}, \mathcal{F}) \\ &= \mathrm{Mor}_{Mod(\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}). \end{aligned}$$

where the second is Lemmas 008U and the first is by Lemma 0089. □